ECO7101

CHAPTER-20

CONSUMER CHOICE: MAXIMIZING UTILITY AND BEHAVIORAL ECONOMICS

Utility is a measure of the satisfaction, happiness, or benefit that results from the consumption of a good or the ability of a good to satisfy your wants.

Total utility is the total satisfaction one receives from consuming a particular quantity of a good. Suppose you purchase 3 units of good X. From 1st unit you get 10 units of utility, from 2nd unit 9 units and from 3rd unit 8 units. Total utility is = 10 + 9 + 8 =27 units.

Marginal utility is the additional utility gained from consuming an additional unit of good X. Marginal utility (MU) is the change in total utility (ΔTU) divided by the change in the quantity (ΔQ) consumed of a good:

$$MU = \frac{\Delta TU}{\Delta Q}$$

Suppose you receive 10 utils of total utility from consuming 1 apple and 19 utils of total utility from consuming 2 apples. Then the marginal utility of the second apple (the additional utility of consuming an additional apple) is 9 (= 19 – 10) utils. As a person consumes more apples, total utility rises, but marginal utility (additional utility received from the additional apple) falls. In other words, total utility rises as marginal utility falls.

Total Utility and Marginal Utility Can Move in Opposite Directions

(1) Number of Apples Consumed	(2) Total Utility (utils)	(3) Marginal Utility (utils)
1	10	10
2	19	9
3	27	8

Clearly, in moving from one to two, to three apples, total utility rises but marginal utility falls.

Law of Diminishing Marginal Utility

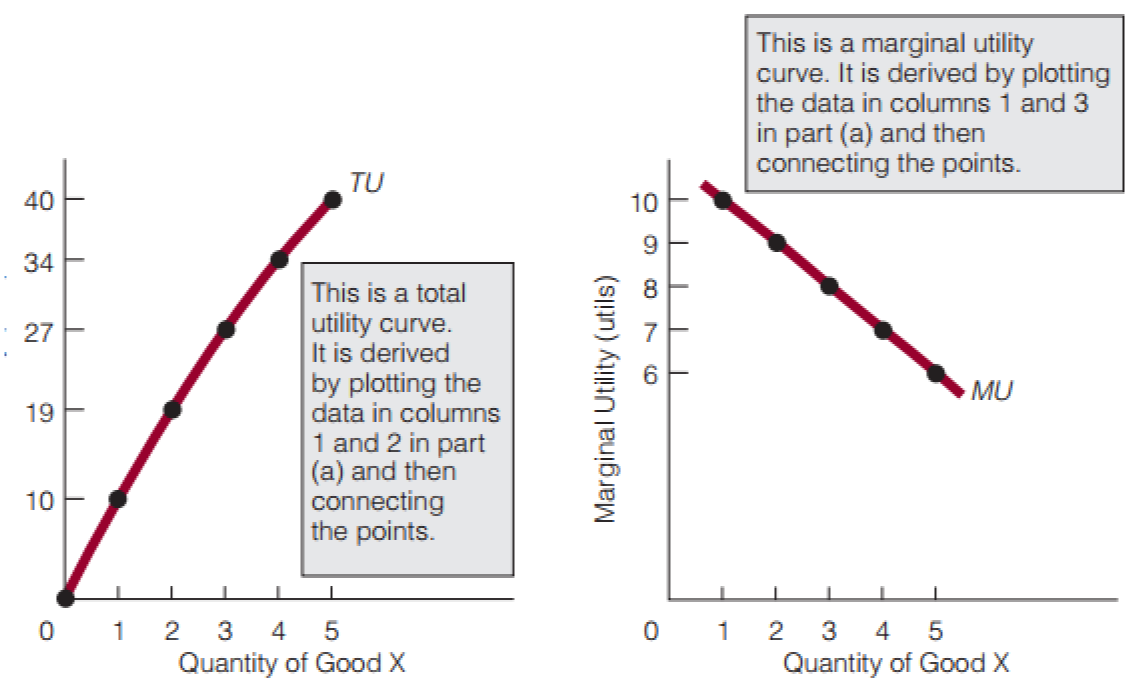
The law of diminishing marginal utility states that, over a given period, the marginal utility gained by consuming equal successive units of a good declines as the amount consumed increases. In other words, the number of utils gained by consuming the first unit of a good is greater than the number of utils gained by consuming the second unit (which is greater than the number gained by the third, which is greater than the number gained by the fourth, and so on).

Exhibit 1

(1) Units of Good X	(2) Total Utility (utils)	(3) Marginal Utility (utils)
0	0	–
1	10	10
2	19	9
3	27	8
4	34	7
5	40	6

(a)

The law of diminishing marginal utility is illustrated in Exhibit 1. The table in part (a) shows both the total utility of consuming a certain number of units of a good and the marginal utility of consuming additional units. Marginal utilities reflect the law of diminishing marginal utility.



(b) **(c)**

In the above figures TU = total utility and MU = marginal utility. In (a) Both total utility and marginal utility are expressed in utils. Marginal utility is the change in total utility divided by the change in the quantity consumed of the good: $MU = \frac{\Delta TU}{\Delta Q}$. (b) Total utility curve. (c) Marginal utility curve. Together, (b) and (c) demonstrate that total utility can increase (b) as marginal utility decreases (c).

The Diamond–Water Paradox

Water is cheap, and diamonds are expensive. But water is necessary to life and diamonds are not. It is odd—even paradoxical—that what is necessary to life is cheap and what is not necessary is expensive? Eighteenth-century economist Adam Smith wondered about this question. He observed that things with the greatest value in use (or that are the most useful) often have a relatively low price and things with little or no value in use have a high price. Smith's observation came to be known as **the diamond–water paradox**, or the paradox of value. The paradox challenged economists, and they sought a solution to it.

Goods have both total utility and marginal utility. Water, for example, is extremely useful: we cannot live without it. Thus, we would expect its total utility (its total usefulness) to be high, but its marginal utility to be low because water is relatively plentiful. As the law of diminishing marginal utility states, the utility of successive units of a good diminishes as consumption of the good increases. In short, water is immensely useful, but there is so much of it that individuals place relatively little value on another unit of it.

In contrast, diamonds are not as useful as water. Hence, we would expect the total utility of diamonds to be lower than that of water, but their marginal utility to be high because there are relatively few diamonds in the world. In other words, the consumption of diamonds (in contrast to that of water) takes place at relatively high marginal utility. Diamonds, which are rare, are used only for their few valuable uses. Water, which is plentiful, gets used for its many valuable uses as well as for its not-so-valuable uses.

So, the total utility of water is high because water is extremely useful. The total utility of diamonds is comparatively low because diamonds are not as useful as water. The marginal utility of water is low because water is so plentiful that people consume it at low marginal utility. The marginal utility of diamonds is high because diamonds are so scarce that people consume them at high marginal utility. **Prices therefore reflect marginal utility, not total utility.**

CONSUMER EQUILIBRIUM: Utility maximization of a consumer in terms Law of Equi-Marginal Utility or Equating Marginal Utilities per Taka.

Suppose there are only two goods in the world: apples and oranges. At present, a consumer is spending his entire income consuming 10 apples and 10 oranges a week. For a particular week, the marginal utility (MU) and price (P) of each are as follows:

$$MU_{oranges} = 30 \text{ utils}$$
$$MU_{apples} = 20 \text{ utils}$$
$$P_{oranges} = \$1$$
$$P_{apples} = \$1$$

So, the consumer's marginal (last) dollar spent on apples returns 20 utils per dollar, and his marginal (last) dollar spent on oranges returns 30 utils per dollar. The ratio MU_O / P_O(O = oranges) is greater than the ratio MU_A/P_A (A = apples).

Here,

$$\frac{MU_O}{P_o} = \frac{30}{1} = 30$$
$$\frac{MU_A}{P_A} = \frac{20}{1} = 20$$

$$Therefore, \frac{MU_O}{P_O} > \frac{MU_A}{P_A}$$

Since oranges give more utility the consumer will buy more oranges and will buy less number of apple as apples give less utility. As a result number of oranges increase that reduces MU_O and as number of apples decreases that increases MU_A. If the process continues MU_O and MU_A will be equal at a point in time. It confirms consumer's equilibrium. The equilibrium that occurs when the consumer has spent all of his or her income and the marginal utilities per dollar spent on each good purchased are equal:

$$\frac{MU_O}{P_o} = \frac{MU_A}{P_A} = \ldots\ldots\ldots$$

A person in consumer equilibrium has maximized total utility. By spending his or her money on goods that give the greatest marginal utility and, in the process, bringing about the consumer equilibrium condition, the consumer is adding as much to total utility as is possible.

Maximizing Utility and the Law of Demand

Suppose a consumer of oranges and apples is currently in equilibrium; that is,

$$\frac{MU_O}{P_O} = \frac{MU_A}{P_A}$$

When in equilibrium, the consumer is maximizing utility. Now, suppose the price of oranges falls. Then the situation becomes

$$\frac{MU_O}{P_O} > \frac{MU_A}{P_A}$$

The consumer will attempt to restore equilibrium by buying more oranges. This behavior—buying more oranges when their price falls—is consistent with the law of demand.

Therefore, the consumer's attempt to reach equilibrium—which is another way of saying that the consumer is seeking to maximize utility—is consistent with the law of demand. In other words, maximization of utility is consistent with the law of demand.

INDIFFERENCE CURVES

Indifference Set: A group of bundles of two goods that give an individual equal total utility.

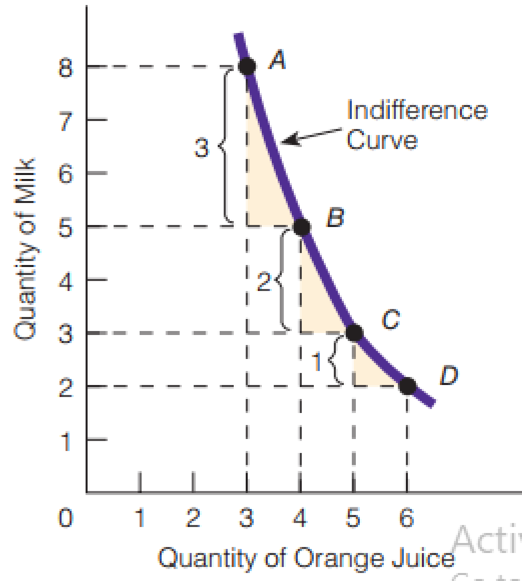
Indifference Curve: The curve that represents an indifference set and that shows all the bundles of two goods giving an individual equal total utility.

A person can be indifferent between two bundles of goods. Suppose that bundle A consists of 2 pairs of shoes and 6 shirts and that bundle B consists of 3 pairs of shoes and 4 shirts. A person who is indifferent between these two bundles is implicitly saying that one bundle is as good as the other, because she receives equal total utility from the two bundles. That implies, at any point on an IC total utility is same irrespective of bundles of commodities. If not, she would prefer one bundle to the other.

If we tabulate all the different bundles from which the individual receives equal utility, we have an indifference set. We can then plot the data in the indifference set and draw an indifference curve. Consider the indifference set in Exhibit 3(a). Of the four bundles of goods, A–D, each bundle gives the same total utility as every other one. These equal-utility bundles are plotted in Exhibit 3(b). Connecting these bundles in a two-dimensional space gives us an indifference curve.

An Indifference Set		
Bundle	Milk (units)	Orange Juice (units)
A	8	3
B	5	4
C	3	5
D	2	6

(a)



(b)

An indifference set is a number of bundles of two goods such that each bundle yields the same total utility. An indifference curve represents an indifference set. In the above exhibit 3, data from the indifference set (a) are used to derive an indifference curve (b).

Characteristics of Indifference Curves

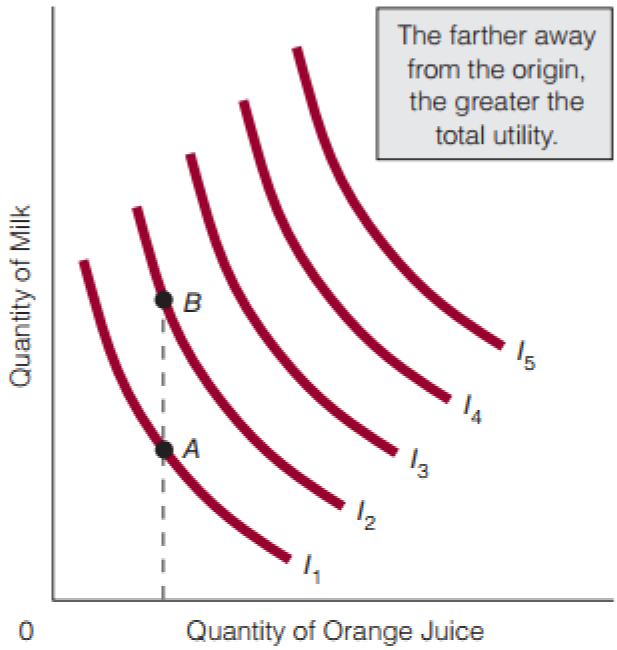
1. **Indifference curves are downward sloping (from left to right).**
   The assumption that consumers always prefer more of a good to less requires that indifference curves slope downward from left to right. More simply, indifference curves are downward sloping because a person has to get more of one good in order to maintain the same level of satisfaction (utility) when giving up some of another good.

2. **Indifference curves are convex to the origin.** As we move down and to the right along the indifference curve, it becomes flatter. the absolute value of the slope of the indifference curve—the marginal rate of substitution—represents the ratio of the marginal utility of the good on the horizontal axis to the marginal utility of the good on the vertical axis:

   $$\frac{MU_{\text{good on horizontal axis}}}{MU_{\text{good on vertical axis}}}$$

   The absolute value of the slope of the indifference curve is the marginal rate of substitution (MRS). The MRS is the amount of one good an individual is willing to give up to obtain an additional unit of another good and maintain equal total utility. Since MRS is negative or MRS declines that's why an IC is convex to the origin.

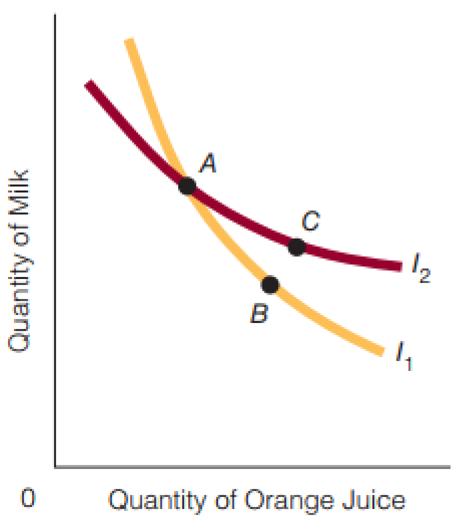
3. **Higher IC gives higher utility**



Higher IC contains more goods. More goods means more utility.
A few of the many possible indifference curves are shown. Any point in the two-dimensional space is on an indifference curve. Indifference curves farther away from the origin represent greater total utility than those closer to the origin.

Indifference Curve Map: A map that represents a number of indifference curves for a given individual with reference to two goods.

4. Two ICs never intersect



4. Suppose two ICs, I_1 and I_2 (where I_1 is lower IC and I_2 is higher IC) intersect at A. It Implies that the higher IC I_2 and lower IC I_1 give same utility at A. It cannot be as it violates transitivity assumption or violates property 3 (Higher IC gives higher utility).

Transitivity

If A is preferred to B and B is preferred to C then A is preferred to C.

BUDGET CONSTRAINT

Budget Constraint: All the combinations, or bundles, of two goods a person can purchase, given a certain money income and prices for the two goods.

Budget Constraint can also be expressed by the following equation known as budget equation.

$M = P_1X_1 + P_2X_2$

Lets solve it for X_2

$M = P_1X_1 + P_2X_2$

$\Rightarrow P_2X_2 = M - P_1X_1$

$\Rightarrow X_2 = \frac{M}{P_2} - \frac{P_1}{P_2}x_1$

M = money income

X_1 = commodity 1

P_1 = Price of commodity 1

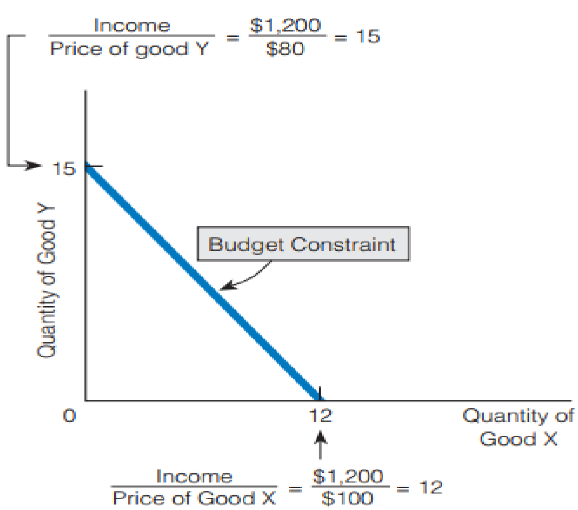
X_2 = commodity 2

P_1X_1 = expenditure on X_1

P_2X_2 - expenditure on X_2

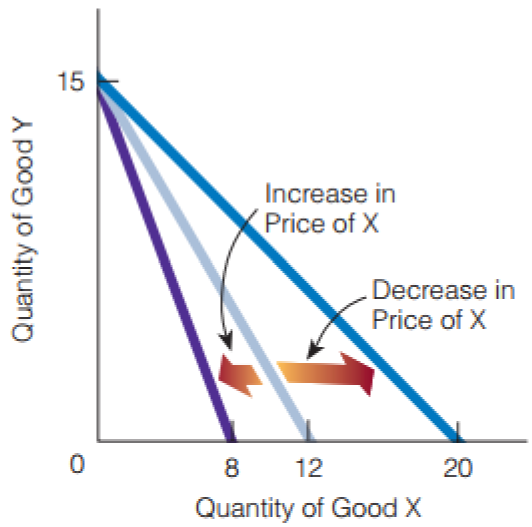
$\frac{M}{P_2}$ = Purchasing power or real income

$\frac{p_1}{p_2}$ = relative price or slope of the budget line

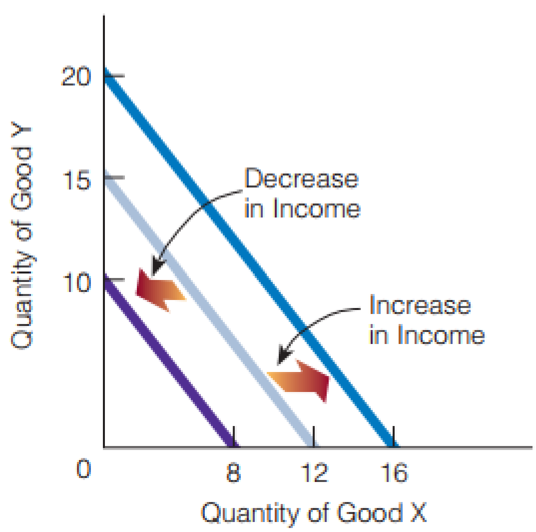


Suppose income of a consumer is Tk. 1200. Suppose price of X_1 commodity is Tk. 100 and price X_2 is Tk. 80. If the consumer spend all his money on either X_1 or X_2, consumer can purchase either 15 units of X_2 or 12 units of X_1. By combining these two extreme points we can derive budget line.

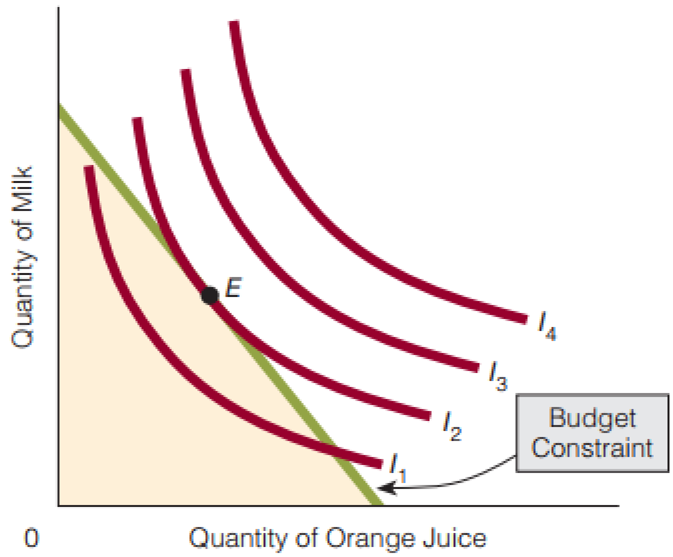
Budget line rotates if prices change or relative price (P_1/P_2) changes or the slope of the budget line changes but real income remains unchanged.



Budget line shifts if real income changes but relative price (P_1/P_2) or the slope of the budget remains unchanged.



Consumer equilibrium or Utility maximization of a consumer in terms IC analysis



Consumer equilibrium exists at the point where the slope of the budget constraint is equal to the slope of an indifference curve or where the budget constraint is tangent to an indifference curve. In the exhibit, this point is E. Here, $\frac{P_{x_1}}{P_{x_2}} = \frac{MU_{x_1}}{MU_{c_2}}$ or, rearranging, we have $\frac{MU_{x_1}}{P_{x_1}} = \frac{MUx_2}{P_{x_2}}$.

The absolute value of the slope of the indifference curve at any point is the marginal rate of substitution, which is equal to the marginal utility of one good divided by the marginal utility of another good, or $\frac{MU_{x_1}}{MU_{x_2}}$.

The necessary condition for consumer equilibrium is obviously that the individual will try to reach a point on the highest indifference curve possible. This point is where the slope of the budget constraint is equal to the slope of an indifference curve (or where the budget constraint is tangent to an indifference curve). At that point, consumer equilibrium is established and the following condition holds:

$$\frac{P_X}{P_Y} = \frac{MU_X}{MU_Y}$$

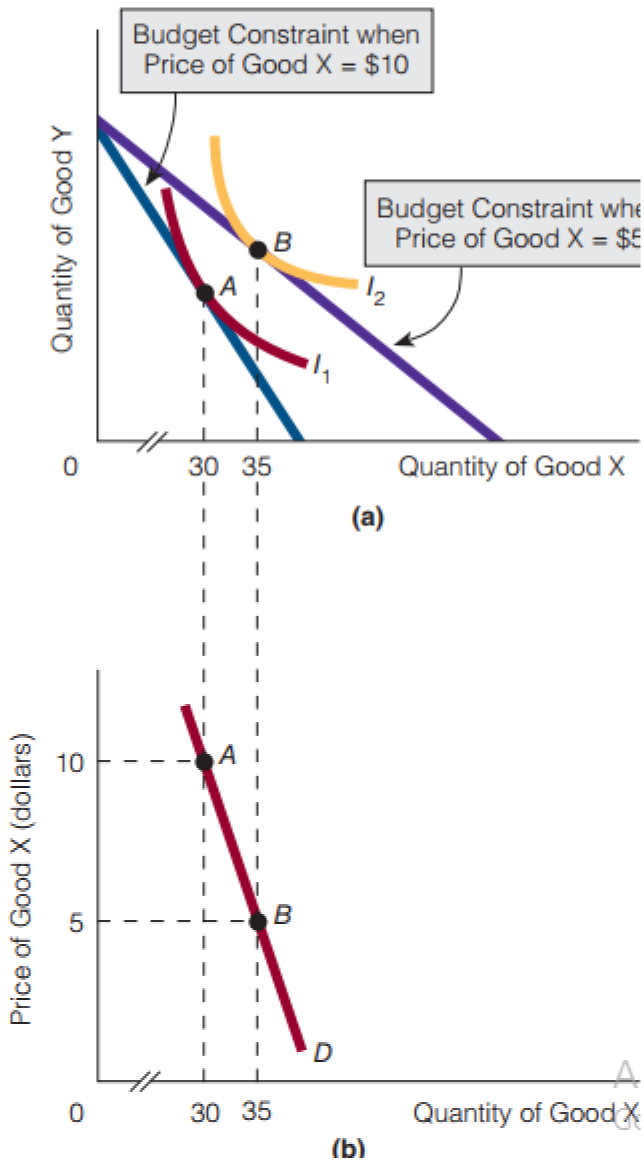
In the figure, the preceding condition is met at point E.

Derivation of demand curve from price consumption curve

We can now derive a demand curve within a budget constraint–indifference curve framework.

Exhibit 8(a) shows two budget constraints, one reflecting a $10 price for good X and the other reflecting a $5 price for good X. As the price of X falls, the consumer moves from point A to point B. At B, 35 units of X are consumed; at A, 30 units of X are consumed. So, a lower price for X results in greater consumption of X. By plotting the relevant price and quantity data, we derive a demand curve for good X in Exhibit 8(b).

Exhibit 8



From Indifference Curves to a Demand Curve (a) At a price of $10 for good X, consumer equilibrium is at point A, with the individual consuming 30 units of X. As the price falls to $5, the budget constraint moves outward (away from the origin) and the consumer moves to point B and consumes 35 units of X. Plotting the price–quantity data for X gives a demand curve for X in (b).

Sample questions for Quiz

What is utility?

Utility is a measure of the satisfaction, happiness, or benefit that results from the consumption of a good or the ability of a good to satisfy your wants.

What is marginal utility?

Marginal utility is the additional utility gained from consuming an additional unit of good X. Marginal utility (MU) is the change in total utility (ΔTU) divided by the change in the quantity (ΔQ) consumed of a good:

$$MU = \frac{\Delta TU}{\Delta Q}$$

What do you mean by the law of diminishing marginal utility?

The law of diminishing marginal utility states that, over a given period, the marginal utility gained by consuming equal successive units of a good declines as the amount consumed increases. In other words, the number of utils gained by consuming the first unit of a good is greater than the number of utils gained by consuming the second unit (which is greater than the number gained by the third, which is greater than the number gained by the fourth, and so on).

What do you mean by the Diamond–Water Paradox?

Water is cheap, and diamonds are expensive. But water is necessary to life and diamonds are not. It is odd—even paradoxical—that what is necessary to life is cheap and what is not necessary is expensive?

What is the condition of maximizing utility of a consumer in terms of law of equi-marginal utility?

$$\frac{MU_O}{P_O} = \frac{MU_A}{P_A}$$

What does the following inequality imply?

$$\frac{MU_O}{P_O} > \frac{MU_A}{P_A}$$

Since oranges give more utility the consumer will buy more oranges and will buy less number of apples as apples give less utility.

Define indifference Curve
The curve that represents an indifference set and that shows all the bundles of two goods giving an individual equal total utility.

What is budget constraint?
All the combinations, or bundles, of two goods a person can purchase, given a certain money income and prices for the two goods.

What are the properties of an indifference curve?
1. Indifference curves are downward sloping 2. Indifference curves are convex to the origin 3. Higher IC gives higher utility 4. Two ICs never intersect

Why an indifference curve is downward sloping?
An indifference curve is downward sloping because a person has to get more of one good in order to maintain the same level of satisfaction (utility) when giving up some of another good.

Why an indifference curve is convex to the origin?
Since MRS is negative or MRS declines that's why an IC is convex to the origin.

Why two indifference curves never intersect?
It cannot be as it violates transitivity assumption or violates property 3 (Higher IC gives higher utility).

Why does budget line rotate?
Budget line rotates if prices change or relative price (P_1/P_2) changes or the slope of the budget line changes but real income remains unchanged.

Why does budget line shift?

Budget line shifts if real income changes but relative price (P_1/P_2) or the slope of the budget remains unchanged.

What is the condition of maximizing utility of a consumer in terms of indifference curve analysis?

$$\frac{P_X}{P_Y} = \frac{MU_X}{MU_Y}$$